

Can I see your ID?

Semantics of Identifier Assignment in BFO/CCO-Aligned Object-Based Intelligence Systems

Forrest Hare (and friends)

Founder, Summit Knowledge Knowledge Solutions LLC

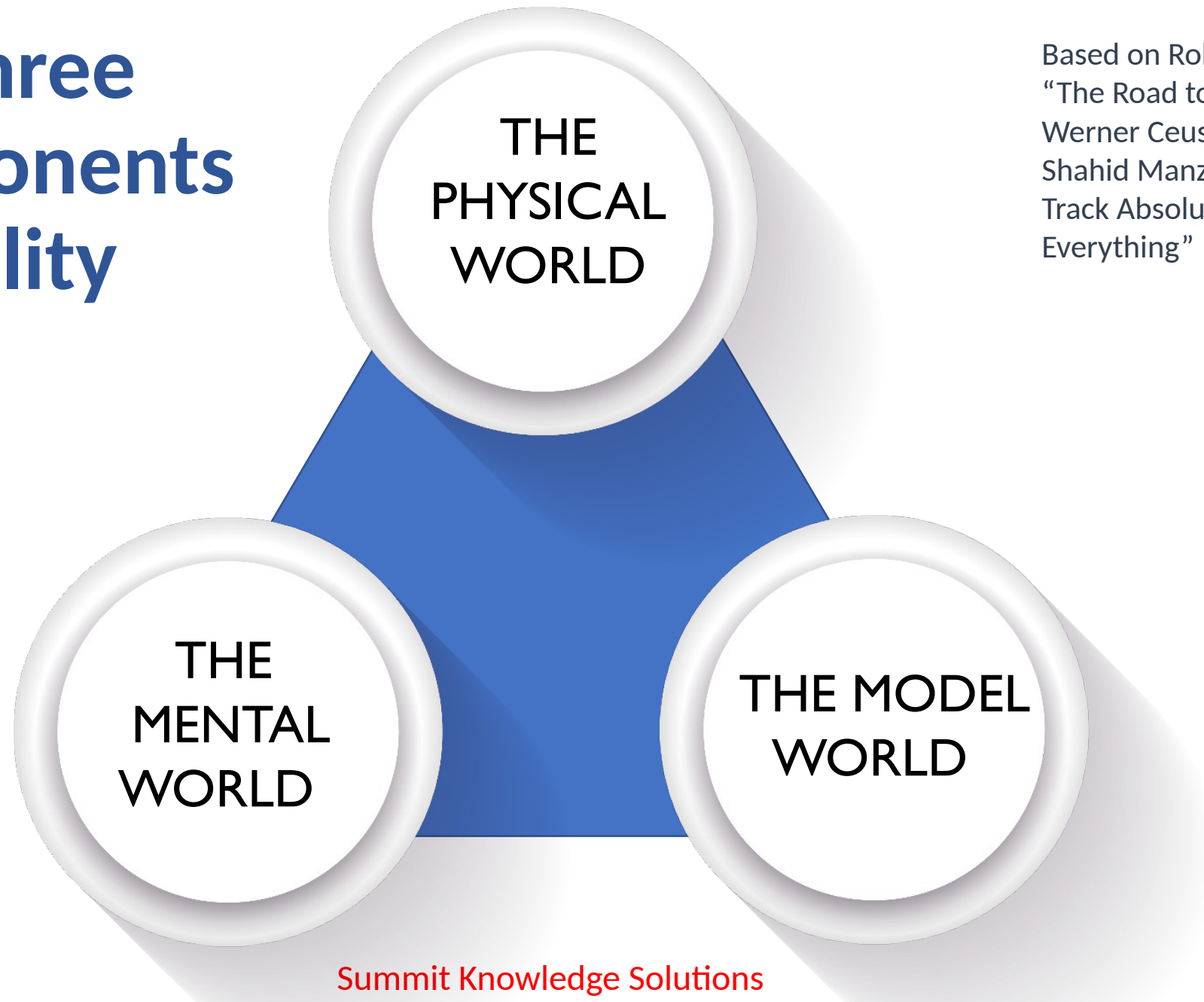
Adjunct Faculty, Schar School of Policy and Government



Overview

- Aligning Worlds and Intelligence
- The Identity Problem in Object-Based Intelligence
- Enterprise Object IDs
- Non-specific Object IDs
- Entity Resolution

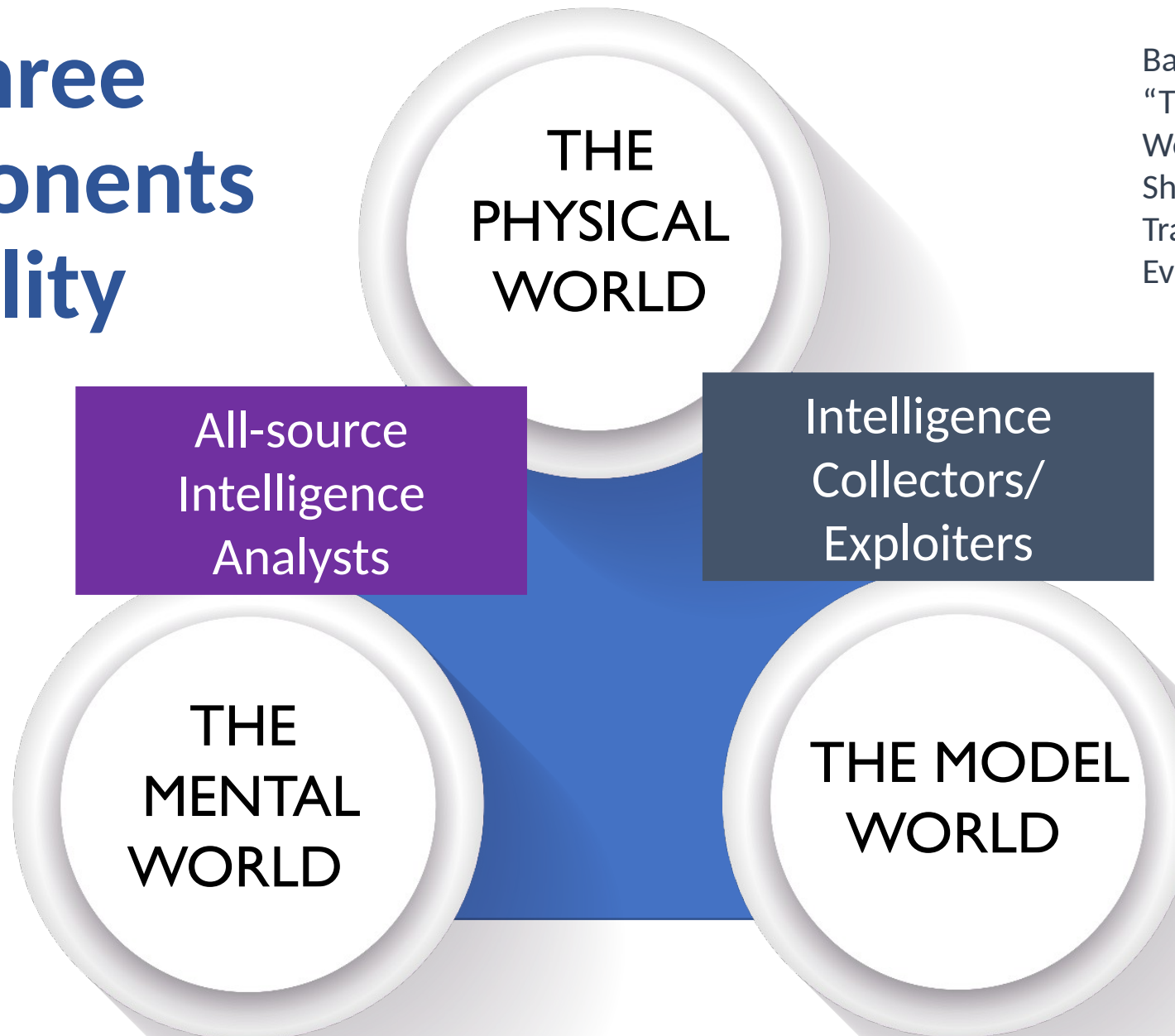
The Three Components of Reality



Based on Robert Penrose, "The Road to Reality" and Werner Ceusters and Shahid Manzoor, "How to Track Absolutely Everything"

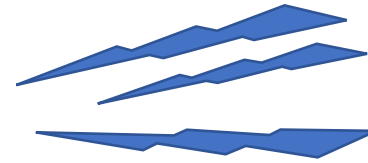
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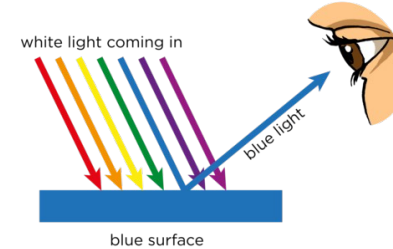


The Physical World

The material entities, and their signatures that can be observed, all existing in the physical world*



Electro-magnetic emissions



Light reflections

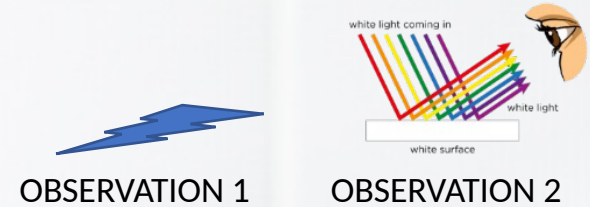


* Our sensors exist in the physical world too, but we will focus on them later

The Mental World

Our perception of the material entities, and the meaning of signatures that can be observed, all existing in the mental world

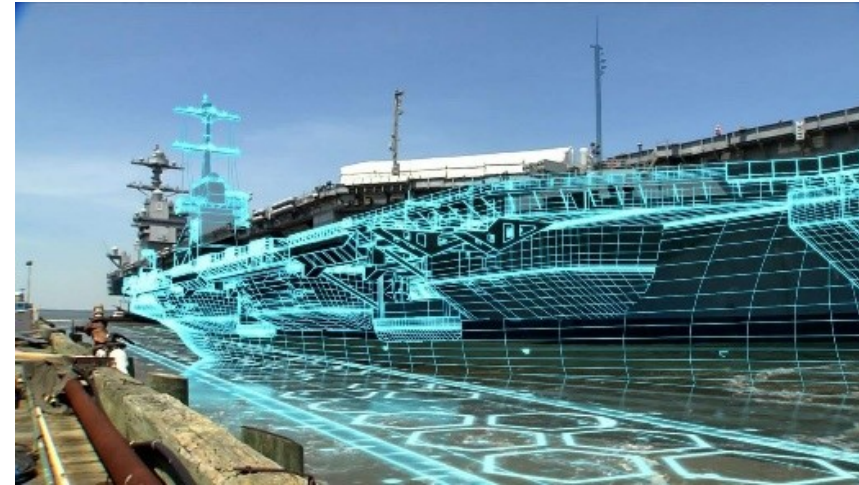
Based on my knowledge (interpretation of the observations), I think there are some signatures in this collection report indicating a tugboat is located at this point at this time. I need to share this information with other analysts.



The Model World

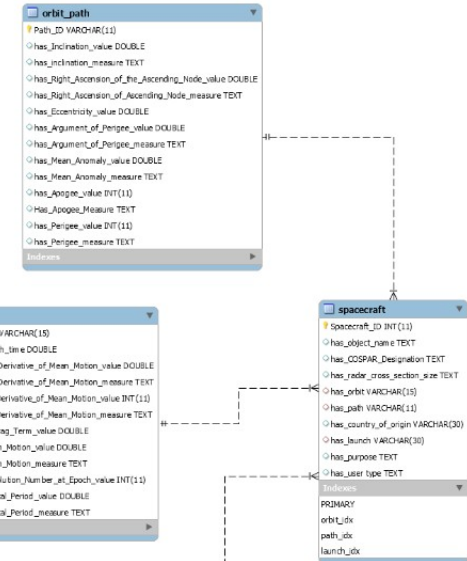
Third-order reality-- created to share a representation of and an understanding of the physical world between us

- ▶ The model of the **knowledge**
 - Formal representation of the meaning of “Tugboat” and “Naval Unit”
- ▶ The model of the **data** that we observe to represent the knowledge
 - Standardized measurement of data related to a tugboat
 - Hull Number (represented as a string in w3C accepted format)
 - Target location (represented as geocoordinate in W3C standardized format)
- ▶ Recorded data is organized in a data store according to **data model** and linked to the **knowledge model**

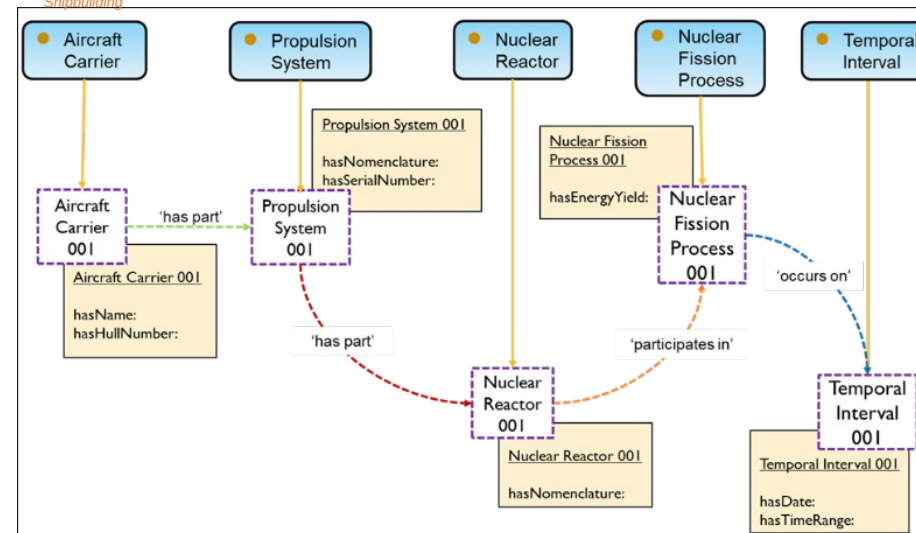


Credit: Newport News Shipbuilding

DATA MODEL

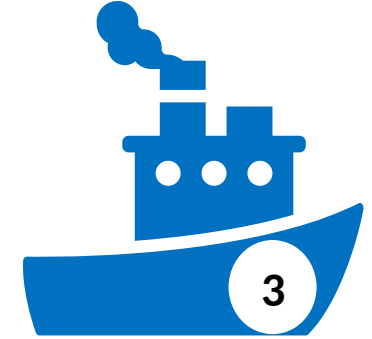
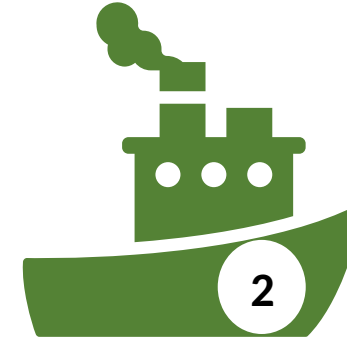
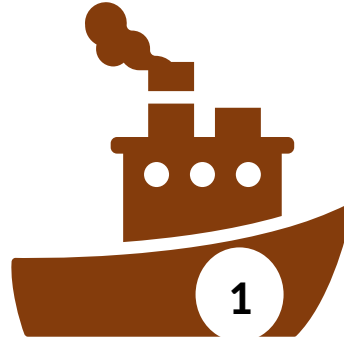


KNOWLEDGE MODEL



All-Source Object Tracking

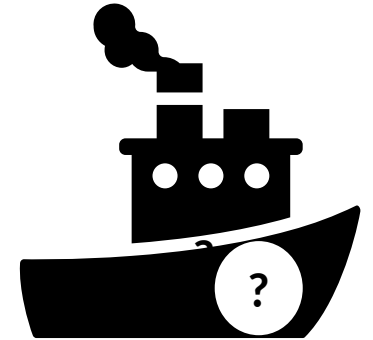
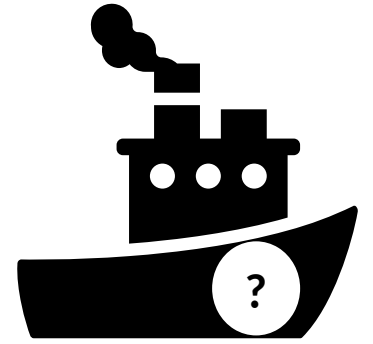
The Oceania Navy has a fleet of tugboats



- ▶ Customers want to know:
 - How many tugboats do they have?
 - Where are they today?
 - What are they doing and why?

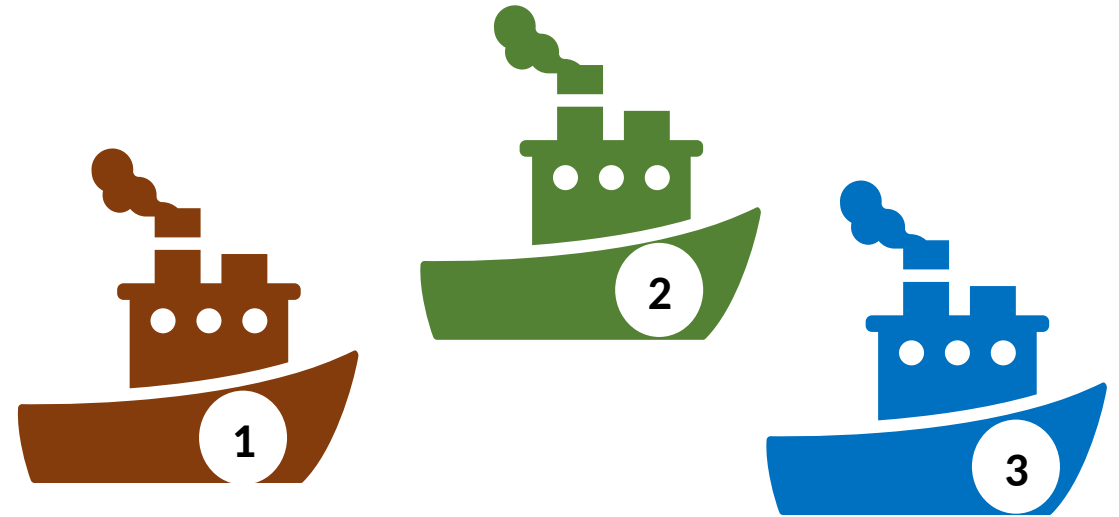
The Object Tracking Challenge

- ▶ Objects move around without letting us know
- ▶ Tracking done by a federation of collectors and all-source analysts
- ▶ Collectors observe signatures of the objects that are ambiguous or not unique to an object
- ▶ Collection occurs over time while objects move
- ▶ All-source analysts need to resolve the observations to the correct tugboats
 - ensure the database most accurately represents reality (to include accurate numbers)



The Enterprise Object ID

- ▶ *An identifier assigned to an object when the enterprise has sufficient knowledge to track that object as a single, definite individual across all existing and future collection, analysis, and reporting*
- ▶ Value must be universally unique and should hold no information
- ▶ One, and only one assigned to each tugboat in the Navy
- ▶ It is an error if:
 - Two objects have the same EOID
 - One object has two EOIDs
- ▶ Requires centralized management

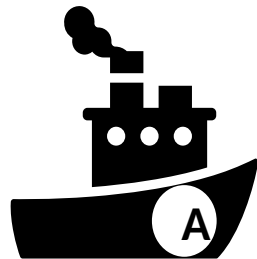
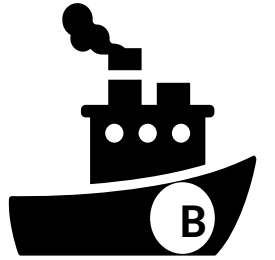


Eoids For the Philosophers

- ▶ Rigidly signifies existence and singularity (Kripke)
 - Semantic content is exhausted by the object it denotes
 - The same continuant (and occurrent) in every scenario in which it exists
- ▶ Linguistic handle for object's haecceity: identical to that individual
- ▶ If an EOID-designated vessel changes any identifiers, the EOID continues to designate the same vessel
- ▶ *Systems that conflate identifier identity with a descriptive signature (e.g., a hash of selected attributes) will see their identifiers break when those attributes change, which is semantically a failure to designate rigidly.*

The Non-Specific Object ID

- ▶ *An identifier assigned to an observed object, or provisional object hypothesis, when the enterprise has insufficient knowledge to track the object as a definite, known individual.*
- ▶ Context-bound and description-dependent
 - "The tugboat we saw on Wednesday in the harbor"
- ▶ Serves as an ID to organize intelligence around the referent object without requiring object singularity knowledge
- ▶ Does not carry the same strict uniqueness requirements as an EOID; value can come from other non-EOID object IDs



NSOIDs For the Philosophers

▶ Test of an NSOID

- *The ID is an NSOID if the enterprise would regard subsequent discovery that it refers to the same individual as some other NSOID as refinement of knowledge, not an error*

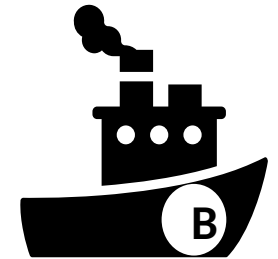
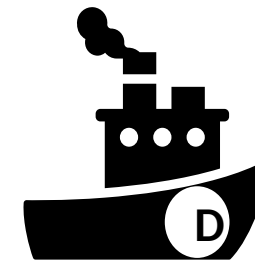
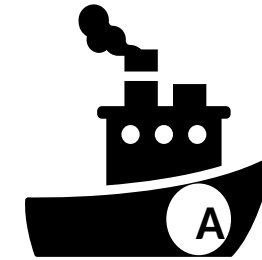
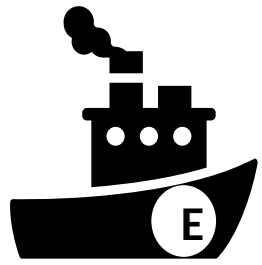
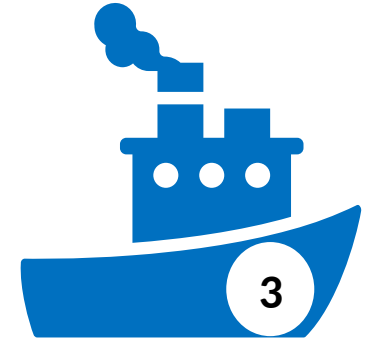
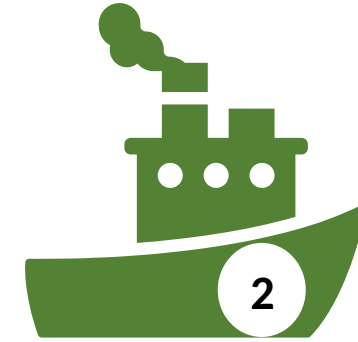
▶ Narrow scope in Russell's sense

- "the system is tracking whatever object matches these properties in this context," rather than "there is a specific, already catalogued, object the system is tracking, and it has these properties."

▶ Narrow scope in Russell's sense

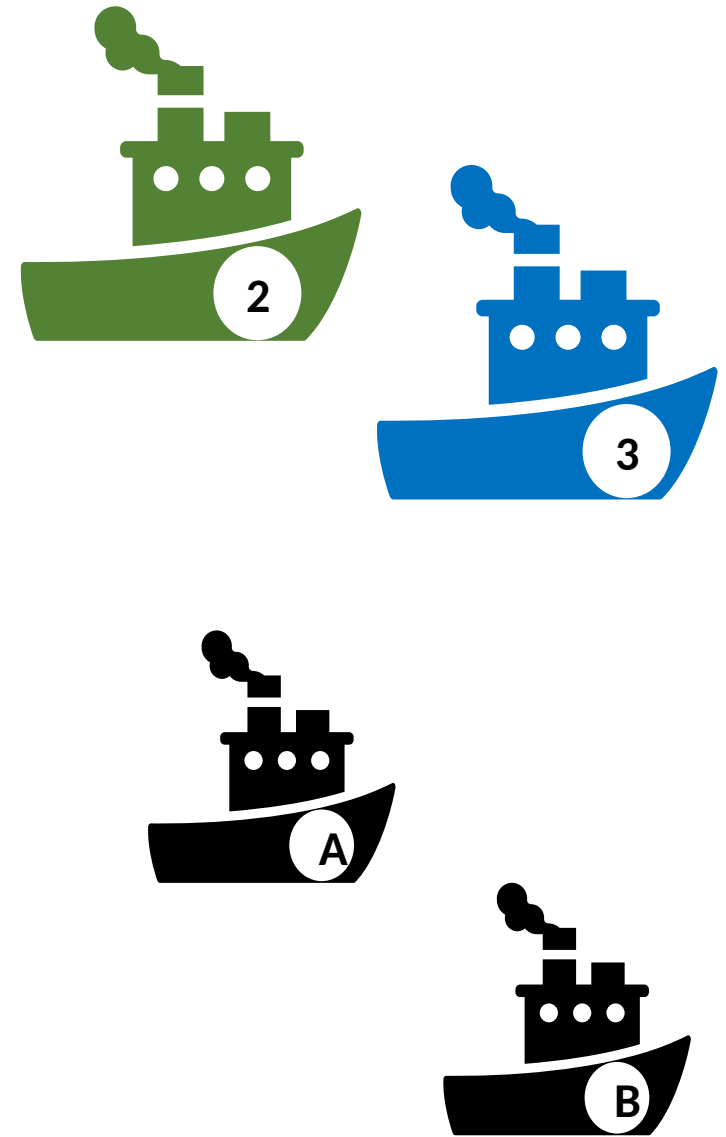
Reality is Ambiguous but Questions Remain

- ▶ Customers want to know:
 - How many tugboats do they have?
 - Where are they today?
 - What are they doing and why?
- ▶ Requires dealing with lots of "dots on the map"
- ▶ Entity resolution process must be governed across the enterprise
- ▶ Must be able to answer the same questions in face of uncertainty



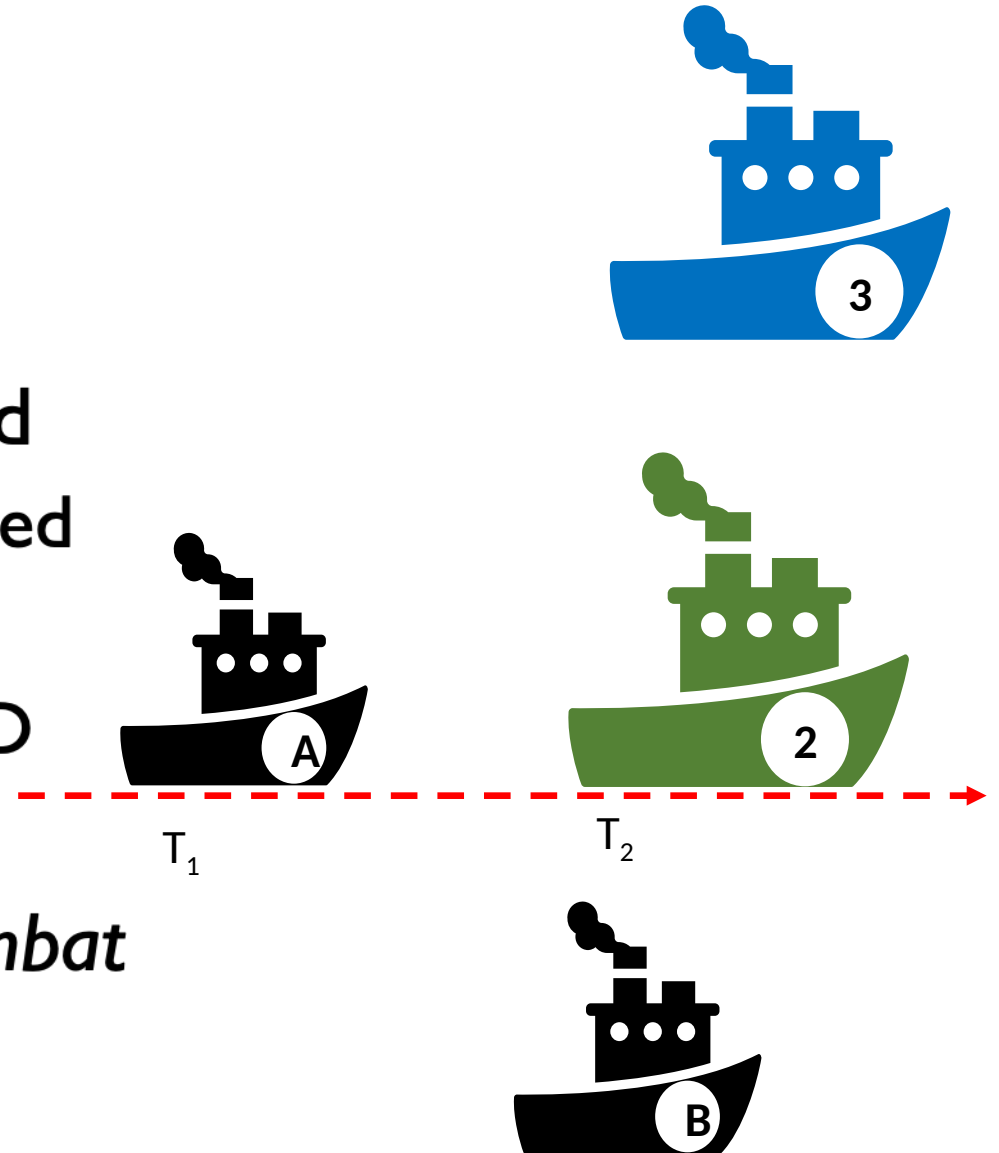
Resolving NSOID Ambiguity

- ▶ Multiple NSOIDs for same individual can collapse to one
 - Occurs when short of resolution threshold
 - Observational history carries forward
- ▶ Not a "data hygiene" cleanup
- ▶ A co-reference discovery ($A \rightarrow B$)
- ▶ Can be reversed at any time (A and B)
- ▶ *Complex process during crises and combat*



Resolving to EOIDs

- ▶ Most consequential process
- ▶ Provisional ID → Rigid ID
- ▶ A “warranted process” and recorded
 - Not a correctness statement, but vetted
 - Follows intel tradecraft standards
 - Multiple outcomes regarding the EOID
- ▶ Must be reversible
- ▶ *Complex process during crises and combat*



Fun with SPARQL-1

- *Audit trail for active EOIDs: retrieve every resolution act that introduced a still-active EOID, with antecedent NSOID, agent, and evidential grounds*

```
PREFIX ri: <http://example.org/ri/>
PREFIX cco: <https://www.commoncoreontologies.org/>
SELECT ?eoid ?antecedentNsoid ?act ?agent ?evidence
WHERE {
  ?eoid a ri:EOID .
  ?act a          ri:ResolutionAct ;
       ri:introduces      ?eoid ;
       ri:hasAntecedent   ?antecedentNsoid ;
       cco:ont00001833    ?agent ;      # has_agent
       ri:hasEvidentialGround ?evidence .
  FILTER NOT EXISTS { ?inv a ri:InvalidationAct ; ri:invalidates ?eoid . }
}
```

*Note1: EOID and NSOIDs
are cco: Designative ICEs*

*Note2: Easier to do this
with RDF Star*

Fun with SPARQL-2

- *Federated co-reference claims with confidence below an audit threshold: the query for reviewing the integrity of cross-enterprise identity claims*

```
PREFIX ri: <http://example.org/ri/>
PREFIX cco: <https://www.commoncoreontologies.org/>
SELECT ?eoidA ?eoidB ?claim ?confidence ?agent
WHERE {
    ?claim a          ri:CoReferenceClaim ;
           ri:between  ?eoidA ;
           ri:between  ?eoidB ;
           ri:hasConfidence ?confidence ;
           cco:ont00001833 ?agent .
    ?eoidA a ri:Eoid .
    ?eoidB a ri:Eoid .
    FILTER ( STR(?eoidA) < STR(?eoidB) )
    FILTER ( ?confidence < 0.85 )
}
```

*Note1: EOID and NSOIDs
are cco: Designative ICEs*

*Note2: Easier to do this
with RDF Star*

Related and Future Work

- Referent Tracking
 - Ceusters and Smith's RT paradigm direct antecedent
- Future Work
 - Testing the concepts in production intelligence systems at scale
 - Most-efficient expressions for provenance (e.g. rdf star)

Discussion

